

*MA.8.AR.4.3***Benchmark**

MA.8.AR.4.3 **Given a mathematical or real-world context, solve systems of two linear equations by graphing.**

Benchmark Clarifications:

Clarification 1: Instruction includes approximating non-integer solutions.

Clarification 2: Within this benchmark, it is the expectation to represent systems of linear equations in slope-intercept form only.

Clarification 3: Instruction includes recognizing that parallel lines have the same slope.

Connecting Benchmarks/Horizontal Alignment

- MA.8.AR.3.4, MA.8.AR.3.5
- MA.8.AR.4.1, MA.8.AR.4.2

Terms from the K-12 Glossary

- Linear Equation

Vertical Alignment**Previous Benchmarks**

- MA.7.AR.4.3, MA.7.AR.4.4, MA.7.AR.4.5

Next Benchmarks

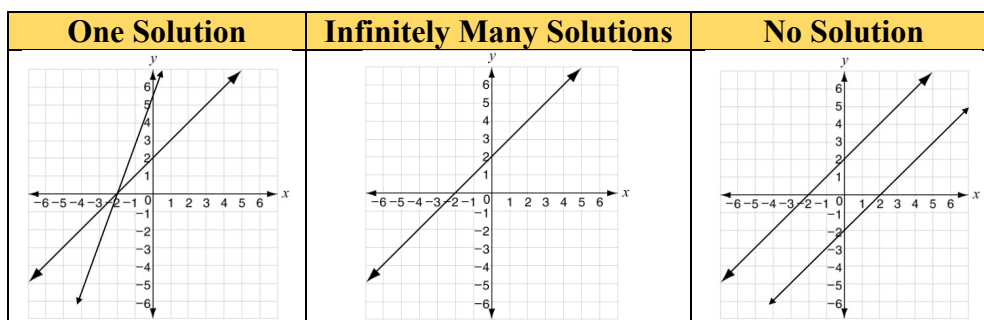
- MA.912.AR.9.1, MA.912.AR.9.6

Purpose and Instructional Strategies

In grade 7, students determined constants of proportionality and graphed proportional relationships from a table, equation or a written description given in a mathematical or real-world context. In grade 8, students extend this learning to a system of two linear equations and graphing the system on the same coordinate plane then students may determine whether there is one solution, no solution or infinitely many solutions. In Algebra 1, students will write and solve a system of two-variable linear equations algebraically and graphically given a mathematical or real-world context.

- Systems of linear equations can have one solution, infinitely many solutions or no solutions.
 - A system of linear equations whose graphs meet at one point (intersecting lines) has only one solution, the ordered pair representing the point of intersection.
 - A system of linear equations whose graphs are coincident (the same line) has infinitely many solutions, the set of ordered pairs representing all the points on the line.

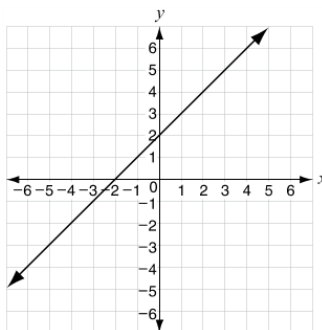
- A system of linear equations whose graphs do not meet (parallel lines) has no solutions and the slopes of these lines are the same. The technical name for these kinds of systems is "inconsistent."



- A system of linear equations is two linear equations that should be solved at the same time. Instruction includes understanding that systems are on the same coordinate plane to determine solutions (*MTR.4.1*).
- The purpose of this benchmark is to focus on graphing to solve the system of equations. This allows for the visual representation of what the solution means in context (*MTR.7.1*).
- Instruction includes recognizing when the system does not have a solution: if there are two distinct lines, but the slopes of the two lines are the same, then the result is a pair of parallel lines. This could be modeled on a graph on paper or through an online resource to support students being able to visualize the lines.

Common Misconceptions or Errors

- Students make errors in plotting points and graphing lines on the coordinate plane, leading to incorrect solutions. To address this misconception, use graph paper, a printed coordinate plane or an online tool for graphing.
- Students incorrectly identify the solution to equations of the same line by stating only the graphed points are the solution set.
 - For example, in the system below with the infinitely many solutions, students may incorrectly not identify (7, 9) as a solution because it is not a point graphed on the coordinate plane.



Strategies to Support Tiered Instruction

- Instruction includes the use of graph paper, a printed coordinate plane, or an online tool for graphing.
- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions.
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Instruction includes drawing connections between systems of equations represented graphically and with equations. Using a graphic organizer, reinforce the solution to a system of equation as the ordered pair that satisfies both equations simultaneously.
 - When there is one solution, the two lines intersect at one point and when using substitution, the coordinates of that one point will result in true statements for both equations.
 - When there is no solution, the two lines do not intersect and therefore there are no coordinates that will result in true statements for both equations.
 - When there are infinite solutions, the two lines coincide and intersect with an infinite number of points. When using substitution, all the points on the lines will result in true statements for both equations.
- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?

Instructional Tasks

Instructional Task 1 (MTR.6.1)

Part A. Graph the line $y = 2x + 2.5$ on a coordinate plane. Draw two other lines with the same slope but different y -intercepts.

Part B. Compare the lines graphed in part A. What do you notice about the other two lines when compared to the given line?

Instructional Items

Instructional Item 1

Solve the system of linear equations by graphing.

$$y = x + 5$$

$$y = 3x - 3$$

Instructional Item 2

Solve the system of linear equations by graphing.

$$y = -\frac{3}{4}x + 5 \qquad y = -4x - 2$$

Instructional Item 3

Solve the system of linear equations by graphing.

$$y = 2x + 6 \qquad y = 2x - 4.2$$

Instructional Item 4

Solve the system of linear equations by graphing.

$$y = -1.5x + 2 \qquad y = 2 - \frac{3}{2}x$$

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive*